

BEIMING LI

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EDUCATION

University of Pennsylvania

PhD, Electrical and Systems Engineering

Master of Science in Engineering, Robotics

Teaching Assistant: Principles of Deep Learning, Learning in Robotics, Graph Neural Networks

Philadelphia, PA

Expected 2029

May 2024

University of Michigan

Bachelor of Science, Computer Engineering

Ann Arbor, MI

Apr. 2022

PUBLICATIONS

Preprints

- **Beiming Li**, Jaime Romero, Jonathan Diller, Vijay Kumar, and Alejandro Ribeiro, “Visual Navigation Transformer with Pose Attention”, *arXiv preprint arXiv:2609.21212*, 2026 [[Preprint](#)][[Code](#)]
- **Beiming Li**, Sergio Rozada, and Alejandro Ribeiro, “Learning Policy Representations for Steerable Behavior Synthesis”, *arXiv preprint arXiv:2601.22350*, 2026 [[Preprint](#)][[Code](#)]

Peer-Reviewed

- Yuezhan Tao, Eran Iceland, **Beiming Li**, Elchanan Zwecher, Uri Heinman, Avraham Cohen, Amir Avni, Oren Gal, Ariel Barel, and Vijay Kumar, “Learning to Explore Indoor Environments using Autonomous Micro Aerial Vehicles”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2024 [[Preprint](#)][[Video](#)]
- Yuezhan Tao, Yuwei Wu, **Beiming Li**, Fernando Cladera, Alex Zhou, Dinesh Thakur, and Vijay Kumar, “SEER: Safe Efficient Exploration for Aerial Robots using Learning to Predict Information Gain”, *IEEE International Conference on Robotics and Automation (ICRA)*, 2023 [[Preprint](#)][[Video](#)][[Code](#)]

RESEARCH EXPERIENCE

University of Pennsylvania

Advisors: Dr. Vijay Kumar, Dr. Alejandro Ribeiro

Philadelphia, PA

Jul. 2022 – Present

Visual Navigation with Pose-Indexed Spatial Memory

Preprint, 2026

- Proposed VNT-PA, a transformer planner that uses a set of pose-indexed depth keyframes as its environment representation instead of an explicit map, trained by imitating a shortest-path expert that plans on the ground-truth scene mesh
- Used camera poses as rotary positional embeddings, which sped up convergence 4× and improved long-horizon navigation over sequence-based baselines
- Reached 93.3% success and 90.4% SPL on HM3D validation scenes and degraded gracefully under localization noise

Policy Representation Learning for Steerable Behavior Synthesis

Preprint, 2026

- Represented policies as expected state-action features under their occupancy measures with a set-based transformer
- Combined variational and contrastive objectives so that the latent space is smooth and policies with similar value functions map to nearby points, allowing gradient-based search in latent space
- Enabled zero-shot behavior synthesis by projected primal-dual optimization over latent codes, steering policies to a target return for one objective subject to unseen constraints on the others, with 98% success in multi-objective MuJoCo

Learning-Based Autonomous Exploration for Aerial Robots

ICRA 2023, ICRA 2024

- Trained a 3D occupancy predictor on partially observed maps from real building scans to estimate the information gain of candidate viewpoints, cutting gain-estimation error by 40% and leading to more efficient exploration
- Coupled a map predictor with a deep RL planner via a prediction-accuracy reward, reached 95% coverage with 25% shorter exploration paths than the state of the art, and deployed the full stack on a quadrotor for indoor flights

Transparent Surface Perception with Polarization Cameras

M.S. Thesis, 2024

- Estimated dense depth for glass surfaces from a single polarization image by segmenting the glass, predicting its surface normal, and extending the surrounding monocular depth across the glass consistent with that normal

SKILLS

Python, C++, PyTorch, ROS/ROS2, Isaac Sim, MuJoCo, Gazebo, ONNX